Cross-fitted stacking with DFT transfer descriptors on the Matbench composition tasks: two pre-registered, fully nested evaluations

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Abstract

We evaluate cross-fitted stacking of composition-based experts on the four composition tasks of Matbench v0.1 (steels, expt_gap, expt_is_metal, glass). The protocol is the same for both runs:

- only the official folds are used;
- every model, hyperparameter and design choice is made by cross-validation inside each training fold;
- the analysis plan was written, and the code frozen by SHA-256 hash, before the test folds were scored.

First pre-registered run. It tested whether a novelty-damped local out-of-fold residual correction, confirmed in earlier chemistry work, transfers to materials. It transferred only weakly: a 0.7% MAE gain on the experimental band gap (5/5 folds, Holm-adjusted $p = 0.005$), and no effect elsewhere.

Second pre-registered run. Written after the first results were seen, and disclosed as such, it strengthened the pipeline with:

- further label-free descriptors;
- a transfer-learning block of DFT (PBE) gaps and formation energies from offline Materials Project data;
- a hurdle treatment of the zero-inflated band gap;

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- Optuna-tuned gradient boosting;
- a nested-CV gate for the local correction, which then kept the correction in none of the 20 folds.

Against the leaderboard leaders, paired over the same test compositions with their public predictions:

- **expt_gap:** our MAE is 0.2352 eV against 0.2865 eV for the best entry ($p_{\text{Holm}} < 0.001$).
- **expt_is_metal:** on par (balanced accuracy 0.9547 vs. 0.9598).
- **steels:** on par (MAE 80.10 vs. 79.95 MPa).
- **glass:** behind (0.8937 vs. 0.9603).

We also show that the leaders of the classification tasks are scored by balanced accuracy, not probability ROC-AUC. Everything ran on one 24-core CPU workstation.

Keywords: materials informatics, composition-based property prediction, Matbench, stacking, residual correction, transfer learning, nested cross-validation, pre-registration

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1. Introduction

Composition-only property models remain the workhorse of materials screening whenever the crystal structure is unknown, and the Matbench v0.1 suite (Dunn et al., 2020) has become the common yardstick for them. Four of its tasks are composition-only and comparatively small: the yield strength of steels (312 alloys), the experimental band gap (4,604 compounds), the experimental metal/non-metal label (4,921) and glass-forming ability (5,680). On these tasks the leaderboard (Matbench developers, 2026) is led by quite different model families: automated machine-learning pipelines on steels, feature-selected neural networks (MODNet, De Breuck et al., 2021b) on glass, and the Darwin entry on the two experimental band-gap tasks. The composition models we are aware of either fit one global predictor or combine predictors with weights that are the same for every composition (as automated-ML ensembles and stacked models do) (De Breuck et al., 2021b; Wang et al., 2021; Goodall and Lee, 2020; Gouvêa et al., 2025; Zou et al., 2025).

Errors of such models are rarely uniform over composition space. Chemical families share systematic biases, and a residual that is large for one alloy is often large for its neighbours. In geostatistics this observation underlies regression kriging, where a trend model is followed by interpolation of its residuals (Hengl et al., 2007); in quantum chemistry, Δ-learning corrects a cheap baseline with a learned residual (Ramakrishnan et al., 2015). In earlier work on chemical property prediction (SIG-Net, Gurram, 2026) we combined a cross-fitted stack of diverse experts (Wolpert, 1992; Breiman, 1996; van der Laan et al., 2007) with a *novelty-damped local out-of-fold (OOF) residual correction*: each prediction is shifted by a kernel-weighted average of the OOF residuals of its nearest training neighbours, and the shift is damped where the query is far from all of them. That correction was confirmed in a pre-registered replication.

Here we ask whether the same construction transfers to composition-based materials property prediction, under the strictest protocol we could devise on a public benchmark: (i) only the official Matbench folds are used; (ii) every choice (expert hyperparameters, stacking weights, correction strength, neighbourhood size, bandwidth, damping and kernel space) is made by cross-validation inside each training fold; (iii) the analysis plan, endpoints and tests were written and the code was frozen by cryptographic hash before any official test fold was predicted; and (iv) every task, fold and arm is reported. We also test a variant suggested by a structured ideation-and-review process (Section 3.5): a *fold-matched* correction that removes a double-counting term which arises because the deployed stack has already been trained on the neighbours whose residuals it borrows.

The first pre-registered run answered that question: the correction transferred only weakly (Section 4), and the pipeline ranked behind the leader on every task. We then wrote a second, disclosed amendment to strengthen the pipeline under the same rules (Section 3.8). It added label-free descriptors, a transfer-learning block of DFT data, a hurdle model for the zero-inflated experimental band gap and tuned gradient boosting, and it scored the test folds once more (Section 5). Both runs are reported in full.

2. Related work

Composition models on Matbench. Magpie descriptors with random forests (Ward et al., 2016) and the Automatminer reference (Dunn et al., 2020) established the baselines; MODNet (De Breuck et al., 2021b,a) added relevance-

redundancy feature selection and joint learning for small data; CrabNet (Wang et al., 2021) and Roost (Goodall and Lee, 2020) learn representations directly from stoichiometry; MatterVial fuses latent, symbolic and descriptor features into a MODNet model (Gouvêa et al., 2025); and tabular foundation models such as TabPFN (Hollmann et al., 2025) are now competitive on small tabular sets. None of these adapts its prediction to the residual structure of nearby training compositions.

Ensembles and stacking for materials. Stacked ensembles have been used for experimental band gaps with DFT-trained first-layer models (Kauwe et al., 2020), for thermodynamic stability from electron-configuration and other composition models (Zou et al., 2025), and for alloy mechanical properties on single datasets with random splits (e.g. Zhao et al., 2025). A mixture of experts over pretrained extractors addresses data scarcity (Chang et al., 2022), and an "ensemble of experts" built from networks pre-trained on related polymer properties was recently proposed in this journal (Miccio, 2025); the latter is transfer learning from SMILES and contains no stacking or residual stage.

Residual correction. Regression kriging (Hengl et al., 2007) and Δ-learning (Ramakrishnan et al., 2015) are the closest classical analogues of a local residual correction. A recent preprint corrects a convex neural stack by global kernel-ridge regression of its residuals in operator learning (Shmalo, 2026). None of these uses OOF residuals with novelty damping in composition space, or evaluates on Matbench folds; the method itself, however, is not new in its components, and our contribution is the pre-registered transfer test, the fold-matched variant and the analysis of where the correction helps or hurts. A structured prior-art check (details in the Supplementary Material) rated the overlap as medium: every component has ancestors, but we found no work that combines them on these tasks.

3. Methods

3.1. Data, splits and features

We use the four composition tasks of Matbench v0.1 through the `matbench` package (`MatbenchBenchmark`), with the official five folds; nothing is re-split. Each composition is described by label-free features computed with `matminer` (Ward et al., 2018a) and `pymatgen` (Ong et al., 2013): 132 Magpie element-property statistics (Ward et al., 2016), 103 element fractions, and 21 further

descriptors (stoichiometric norms, valence-orbital fractions (Meredig et al., 2014), ionic character and charge-balance flags, band centre and atomic-orbital HOMO/LUMO estimates). A handful of compositions (0, 14, 14 and 21 rows in the four tasks) make one of the extra featurisers fail; these values are imputed with the feature's dataset-wide median, which uses no labels.

3.2. Experts and cross-fitting

Inside an official training fold T_k the training rows are split into $J = 5$ inner folds $P_1, \ldots, P_J$ (seed 0; stratified for classification). Every expert e is fitted J times on $T_k \setminus P_m$ to produce OOF predictions $\hat{f}_e^{(-m(i))}(x_i)$, and once on all of T_k to predict the test rows. The expert pool is deliberately heterogeneous and CPU-feasible: LightGBM (Ke et al., 2017) on all features (for expt_gap also an L1-objective copy), random forests (Breiman, 2001), extremely randomised trees (Geurts et al., 2006), kernel ridge regression on element fractions and on standardised Magpie features (RBF kernel, regularisation and width chosen by an internal three-fold grid search), RBF support-vector classifiers on the same two spaces for the classification tasks, and a two-layer perceptron (Pedregosa et al., 2011). Regression predictions are clipped to the training-target range.

3.3. Simplex stack

The global stack is $s(x) = \sum_e w_e \hat{f}_e(x)$ with $w_e \geq 0$ and $\sum_e w_e = 1$, fitted on the OOF predictions. For regression the weights minimise the mean absolute error (a linear programme, matching the Matbench metric); for classification they minimise the Brier score. To obtain OOF *stack* predictions $s^{\mathrm{oof}}(x_i)$ for the training rows, the weights used for inner fold m are fitted without the rows of P_m.

3.4. Novelty-damped local OOF residual correction

With OOF residuals $r_j = y_j - s^{\mathrm{oof}}(x_j)$, the corrected prediction for a query q is

$$\hat{y}(q) = s(q) + \alpha \, \frac{\sum_{j \in \mathcal{N}_k(q)} w_{qj} \, r_j}{\sum_{j \in \mathcal{N}_k(q)} w_{qj} + \kappa}, \qquad w_{qj} = \exp\big(-(d(q,j)/h)^2\big), \tag{1}$$

where $\mathcal{N}_k(q)$ are the k nearest training compositions under a composition distance d (Euclidean on standardised Magpie vectors, or L1 on element fractions). The pseudo-count κ provides the novelty damping: when all neighbours are far, $\sum w$ is small and the correction vanishes. For classification

the shift is applied on the logit scale with probability residuals. The tuple $(d, k, h, \kappa, \alpha)$ is selected on the training rows only, by scoring the corrected OOF predictions (leave-self-out neighbour sets) with the task metric over a fixed grid: $k \in \{5, 10, 20, 40\}$, h a multiple $\{0.5, 1, 2, 4, 8, 16\}$ of the median nearest-neighbour distance, $\kappa \in \{0, 0.1, 0.5, 2\}$, and $\alpha \in \{0.25, 0.5, 0.75, 1\}$ (regression) or $\{0.1, 0.25, 0.5, 1, 2, 4\}$ (classification). The uncorrected stack ($\alpha = 0$) is always a candidate and wins ties, so the procedure may decline to correct.

3.5. Fold-matched correction

The deployed stack $s = F^{\text{full}}$ was trained on the very neighbours j whose OOF residuals Eq. (1) borrows. Writing $F^{(-m)}$ for the stack whose experts excluded inner fold P_m, the naive transfer decomposes as $r_j = [r_j + F^{(-m(j))}(q) - F^{\text{full}}(q)] + [F^{\text{full}}(q) - F^{(-m(j))}(q)]$. The second bracket, the *absorption term*, measures how far the refit has already moved toward the neighbour, and the naive correction counts it a second time; for interpolating experts and near-duplicate neighbours it can be as large as r_j itself. The fold-matched variant transfers only the first bracket, $T_{qj} = r_j + F^{(-m(j))}(q) - F^{\text{full}}(q)$, in place of r_j in Eq. (1). Its selection uses deployment-matched training instances: for a training row $i \in P_m$ and a neighbour $j \in P_{m'}$ ($m' \neq m$) the transfer is computed from the pair-excluded refit $F^{(-m,-m')}$, which saw neither i nor j; neighbours in the same inner fold use r_j directly. This requires $J(J-1)/2 = 10$ extra fits per expert and fold, with the stack weights held fixed. The variant was proposed by a structured ideation pipeline (ResearchStudio IdeaSpark), whose coherence check verified the algebraic decomposition on synthetic instances; we pre-registered it as a secondary arm, not as the primary method. Relative to that proposal we simplified the damping map to the same (α, κ) family as Eq. (1), used the same two distances instead of the element mover's distance, and did not add TabPFN or MODNet experts, for reasons of CPU cost.

3.6. Hard labels for classification

Most classification entries on the Matbench leaderboard report a "ROC-AUC" equal to their balanced accuracy, which is what `matbench` computes when hard labels rather than probabilities are recorded. For a like-for-like comparison we therefore also report balanced accuracy at a threshold chosen inside each training fold (maximising balanced accuracy on the arm's training OOF predictions), alongside the probability ROC-AUC.

3.7. Protocol, pre-registration and statistics

All model and hyperparameter selection was done by cross-validation within each official training fold. Concretely, the candidate design configurations (with or without the 21 extra descriptors; L2 or L1 stacking loss; with or without an L1-objective LightGBM expert) were first explored by inner cross-validation on the training rows of fold 0 only. Then, inside every official training fold T_k, all candidates were cross-fitted and the one with the best cross-fitted OOF score of the uncorrected global stack was kept; because this criterion uses the uncorrected stack, it favours neither correction arm. The correction hyperparameters were then selected inside T_k as described above. The analysis plan, including an amendment that introduced this per-fold selection and was written before any official test fold was scored, and the SHA-256 hashes of all code were committed before the final run; both are provided in the repository and Supplementary Material. (An earlier run under the original plan, in which the design was fixed on fold 0's training rows, was stopped and never scored.)

The primary endpoint, per task, is Δ = mean over the five official folds of the metric of the corrected stack minus that of the uncorrected stack (MAE for regression, probability ROC-AUC for classification). Its test is a paired bootstrap that resamples test points within folds ($B = 10{,}000$), with Holm correction across the four tasks; a task counts as improved only if Δ favours the correction and the Holm-adjusted $p < 0.05$. Because five folds give little power, we also report per-fold differences, the number of folds improved, and the Nadeau–Bengio corrected resampled t-test (Nadeau and Bengio, 2003). Secondary comparisons (fold-matched versus stack and versus naive, stack versus the best single expert selected by inner CV, kernel-space ablations, correction without stacking) and analyses (errors stratified by distance to the nearest training composition, calibration, selected configurations, and a label-permutation control) were also pre-specified.

3.8. Amendment 2: a stronger, still fully nested pipeline

The Amendment-1 results (Section 4) showed the local correction to be a minor ingredient. The pipeline also ranked behind the leaderboard leader on every task. We therefore wrote a second amendment to the analysis plan. It is disclosed as written *after* the Amendment-1 test results had been seen. The amendment changes only the expert pool, the features and the selection/threshold procedure. As before, every choice is made by inner cross-validation inside each official training fold, and the official test folds

were scored exactly once under it, after a new code freeze. The design work for this amendment used only the training rows of fold 0.

Features. In addition to the Amendment-1 features we compute further label-free composition descriptors with `matminer` (Ward et al., 2018a):

- the "deml", "matminer" and "megnet_el" element-property presets;
- Meredig descriptors (Meredig et al., 2014);
- Miedema mixing enthalpies, Yang solid-solution parameters (Yang and Zhang, 2012) and atomic packing efficiency (Laws et al., 2015) (descriptors developed for glass formation, as implemented by Ward et al., 2018b);
- Wen alloy descriptors and the transition-metal fraction;
- Oliynyk element-property statistics (Oliynyk et al., 2016) from the CBFV package.

DFT transfer block. For each composition we look up the Materials Project (Jain et al., 2013) PBE data distributed with `matminer` (`matbench_mp_gap`, joined to `matbench_mp_e_form`) by reduced formula. The block contains:

- the PBE gap, metallicity and formation energy of the lowest-formation-energy polymorph;
- polymorph statistics: minimum, median and maximum gap, metallic fraction and count;
- a match indicator.

It covers 73% of the expt_gap and 73% of the expt_is_metal compositions, 1% of glass and none of steels. This is transfer learning from computed to experimental properties, in the spirit of Kauwe et al. (2020). The block contains no experimental labels, and no pretrained language model is involved.

Experts.

- LightGBM on all features, tuned inside each training fold with Optuna (Akiba et al., 2019) (20 trials, default configuration evaluated first).
- XGBoost (Chen and Guestrin, 2016).

- CatBoost (Prokhorenkova et al., 2018).
- Extremely randomised trees on three feature sets, including element fractions alone (the strongest single model on steels in the design phase).
- The kernel, support-vector, nearest-neighbour and perceptron experts of Amendment 1.
- For the zero-inflated experimental band gap (about half of the compounds have $E_g = 0$), *hurdle* experts. Each pairs a classifier for $E_g > 0$ with a regressor trained on the $E_g > 0$ rows, and combines them as $p\,r$ or as $r \mathbb{1}[p \geq 0.5]$. The latter is the MAE-optimal rule when the conditional distribution is a point mass at zero plus a positive part.

CrabNet and MODNet experts were implemented but dropped, because a single fit exceeded our CPU budget (about 23 and more than 40 minutes, respectively).

Stack and final pipeline. The simplex stack is as in Amendment 1. The local correction is still selected inside the training fold, but it is now kept only if a *nested* check confirms it. For each inner fold, the correction is re-selected on the other folds and applied to the held-out one, and the correction is retained only if this honest OOF score beats the uncorrected stack. In the design phase this gate prevented a correction that the non-nested selection had switched on from degrading steels. The final pipeline is this gated stack. Classification labels use the balanced-accuracy-maximising threshold chosen on training OOF predictions.

Primary endpoint of Amendment 2. The primary endpoint is the final pipeline versus the leaderboard leader on each task, using the board's own metric semantics. For classification this is balanced accuracy, because the leaders either recorded hard labels (Darwin) or were scored by a `matbench` version that thresholded probabilities at 0.5 (MODNet v0.1.12). We use the leaders' public per-sample predictions (`matbench` repository) to form a paired bootstrap over test points within folds, with Holm correction over the four tasks. We claim to beat a leader only if our mean is better and the Holm-adjusted $p < 0.05$.

Table 1: Every arm on the four Matbench composition tasks (official 5-fold CV; mean ± s.d. over folds). Regression: MAE (lower is better); classification: ROC-AUC from probabilities (higher is better). Single-expert rows are means over the folds in which the expert was part of the selected configuration.

Arm	steels MAE (MPa)	expt_gap MAE (eV)	expt_is_metal ROC-AUC	glass ROC-AUC
Best single (inner CV)	83.41 ± 10.86	0.3259 ± 0.0130	0.9819 ± 0.0015	0.9548 ± 0.0097
Equal-weight mean	102.46 ± 20.11	0.4007 ± 0.0242	0.9811 ± 0.0027	0.9458 ± 0.0116
Simplex stack	83.12 ± 10.89	0.3264 ± 0.0139	0.9825 ± 0.0017	0.9567 ± 0.0091
Stack + local corr. (naive)	82.80 ± 11.65	0.3240 ± 0.0140	0.9825 ± 0.0018	0.9567 ± 0.0091
ablation: Magpie space only	83.53 ± 11.30	0.3235 ± 0.0139	0.9825 ± 0.0017	0.9567 ± 0.0091
ablation: fraction space only	82.80 ± 11.65	0.3235 ± 0.0142	0.9825 ± 0.0018	0.9567 ± 0.0091
Stack + fold-matched corr.	82.02 ± 9.12	0.3276 ± 0.0149	0.9826 ± 0.0017	0.9559 ± 0.0094
Best single + local corr.	82.69 ± 11.60	0.3239 ± 0.0133	0.9819 ± 0.0015	0.9548 ± 0.0097
Expert et_all	83.41	0.3259	0.9809	0.9548
Expert krr_frac	160.38	0.5613	–	–
Expert krr_magpie	109.37	0.4840	–	–
Expert lgbm_all	88.00	0.3425	0.9819	0.9519
Expert lgbm_l1_all	93.97	–	–	–
Expert mlp_all	156.17	0.4611	0.9702	0.9055
Expert rf_all	96.62	0.3645	0.9808	0.9516
Expert svc_frac	–	–	0.9472	0.8735
Expert svc_magpie	–	–	0.9566	0.9009

4. Results I: the transferred local correction (Amendment 1)

4.1. All arms on the official folds

Table 1 lists every arm on every task (mean ± standard deviation over the five official folds), and Table 2 the pre-registered tests. The equal-weight mean of the experts is clearly the worst ensemble on all tasks; the simplex stack never loses to it. Against the best single expert chosen by inner cross-validation, the stack is better on glass (+0.20% ROC-AUC, $p_{\text{boot}} = 0.002$, 5/5 folds) and is within noise on the other three tasks (steels -0.35%, expt_gap +0.13%, expt_is_metal +0.06%). The extremely randomised trees expert carries most of the stack weight on the two regression tasks, and LightGBM and extra trees share it on the classification tasks (Table 6).

4.2. Primary endpoint: does the local correction transfer?

The correction improved 1 of the four tasks under the pre-registered rule. On expt_gap the corrected stack reached an MAE of 0.3240 eV against 0.3264 eV for the uncorrected stack ($\Delta = -0.00232$ eV, -0.71%), with every one of the five folds improved and $p_{\text{Holm}} = 0.005$ (Nadeau–Bengio $p = 0.011$).

Table 2: Pre-registered primary test (naive local correction vs. uncorrected stack) and key secondary comparisons. Δ is the mean over folds of the metric difference (MAE for regression, where negative favours A; ROC-AUC for classification, where positive favours A). CI and p_{boot}: paired bootstrap over test points within folds ($B = 10^4$); p_{Holm}: Holm correction over the four primary tests; p_{NB}: Nadeau–Bengio corrected resampled t-test over the five folds.

Task	Comparison (A vs B)	Δ = A−B (rel.)	95% CI	p_{boot}	p_{Holm}	folds A better	p_{NB}
steels	naive local vs stack (primary)	-0.3202 (-0.39%)	[-2.15, +1.5]	0.720	1.000	3/5	0.896
steels	naive local vs best single	-0.6135 (-0.74%)	[-2.63, +1.34]	0.538	–	3/5	0.816
steels	stack vs best single	-0.2933 (-0.35%)	[-1.65, +1.08]	0.660	–	2/5	0.417
steels	fold-matched vs stack	-1.097 (-1.32%)	[-4.32, +2.13]	0.519	–	2/5	0.817
steels	fold-matched vs naive local	-0.7765 (-0.94%)	[-3.45, +1.91]	0.583	–	3/5	0.812
expt_gap	naive local vs stack (primary)	-0.00232 (-0.71%)	[-0.0037, -0.000942]	0.001	0.005	5/5	0.011
expt_gap	naive local vs best single	-0.001895 (-0.58%)	[-0.00391, +0.000155]	0.071	–	4/5	0.108
expt_gap	stack vs best single	+0.0004248 (+0.13%)	[-0.000988, +0.00181]	0.566	–	3/5	0.695
expt_gap	fold-matched vs stack	+0.001241 (+0.38%)	[-0.00145, +0.00418]	0.390	–	2/5	0.516
expt_gap	fold-matched vs naive local	+0.00356 (+1.10%)	[+0.0012, +0.00602]	0.005	–	0/5	0.096
expt_is_metal	naive local vs stack (primary)	-3.305e-06 (-0.00%)	[-4.88e-05, +4.3e-05]	0.899	1.000	0/5	0.541
expt_is_metal	naive local vs best single	+0.0005853 (+0.06%)	[-0.000504, +0.0017]	0.301	–	4/5	0.348
expt_is_metal	stack vs best single	+0.0005886 (+0.06%)	[-0.000507, +0.0017]	0.275	–	4/5	0.345
expt_is_metal	fold-matched vs stack	+0.0001571 (+0.02%)	[-0.000148, +0.00047]	0.320	–	3/5	0.577
expt_is_metal	fold-matched vs naive local	+0.0001604 (+0.02%)	[-0.000128, +0.000458]	0.277	–	3/5	0.566
glass	naive local vs stack (primary)	+0 (+0.00%)	[+0, +0]	1.000	1.000	0/5	1.000
glass	naive local vs best single	+0.001897 (+0.20%)	[+0.000704, +0.00314]	0.002	–	5/5	0.141
glass	stack vs best single	+0.001897 (+0.20%)	[+0.000689, +0.00314]	0.002	–	5/5	0.141
glass	fold-matched vs stack	-0.0008173 (-0.09%)	[-0.00194, +0.000315]	0.153	–	1/5	0.247
glass	fold-matched vs naive local	-0.0008173 (-0.09%)	[-0.00192, +0.000273]	0.149	–	1/5	0.247

The effect is real but small: about 0.7% of the MAE. On steels the change was -0.39% ($p_{\text{Holm}} = 1.000$, 3/5 folds improved), indistinguishable from zero given a fold-to-fold standard deviation of about 10.89 MPa. On the two classification tasks the inner-CV selection almost never switched the correction on (glass: 0/5 folds; expt_is_metal: 1/5), so the corrected and uncorrected stacks are identical or nearly so (Δ ROC-AUC $= -0.00000$ and $+0.00000$). In short, the correction that was confirmed in our chemistry study transfers only weakly: it gives a consistent sub-percent gain on the largest regression task, nothing measurable on steels, and is (correctly) declined by its own selection procedure on the classification tasks.

4.3. Where the correction helps and hurts

Fig. 1 stratifies the test errors by the distance from each test composition to its nearest training composition. On both regression tasks the naive correction *hurt* in the nearest-support quartile (expt_gap +0.9%, steels +6.8% MAE relative to the stack) and helped in the other three (expt_gap -1.0%, -1.1%, -0.9%). That is the pattern the fold-matched analysis predicts: for a near-duplicate neighbour the deployed stack has already absorbed the neigh-

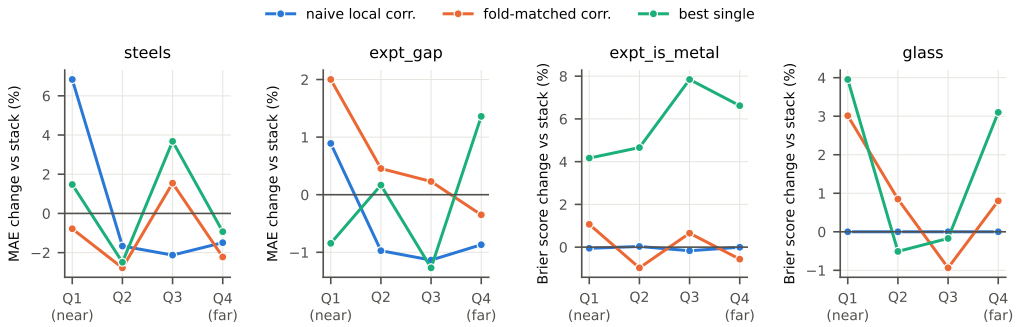


Figure 1: Error change relative to the uncorrected stack, by support quartile (distance from the test composition to its nearest training composition; bin edges are the training rows' leave-one-out quartiles in each fold). Regression: MAE; classification: Brier score. Negative is better.

bour's error, so adding its OOF residual counts it twice. The fold-matched variant, however, did not repair it: on expt_gap it was worse than the naive correction overall (+1.10% MAE, $p_{\text{boot}} = 0.005$, 0/5 folds better) and also worse in the nearest quartile (+2.0%). On steels it gave the lowest mean MAE of all arms (82.02 MPa) but with no detectable difference from the stack (-1.32%, $p_{\text{boot}} = 0.519$). On the classification tasks it was switched on in every fold but changed ROC-AUC by less than 0.1% in either direction.

4.4. Ablations and controls

Restricting the neighbour search to one composition space gave essentially the same gain on expt_gap (Magpie -0.86%, element fractions -0.88%), whereas on steels only the element-fraction space was selected in practice and the Magpie-only variant was slightly worse than the stack (+0.50%, 0/5 folds better). Applying the correction to the best single expert instead of the stack ("correction without stacking") improved expt_gap by -0.63% relative to that expert ($p_{\text{boot}} = 0.002$, 5/5 folds), so on this task the gain does not depend on stacking. The label-permutation control did not behave as a clean null: with labels permuted within inner folds the selection still switched a correction on in every fold of every task (Table 3), because shrinking predictions toward a neighbourhood mean lowers the training loss against permuted labels too. We pre-registered this control as diagnostic only, and it shows that a non-zero selected strength is not by itself evidence of local structure.

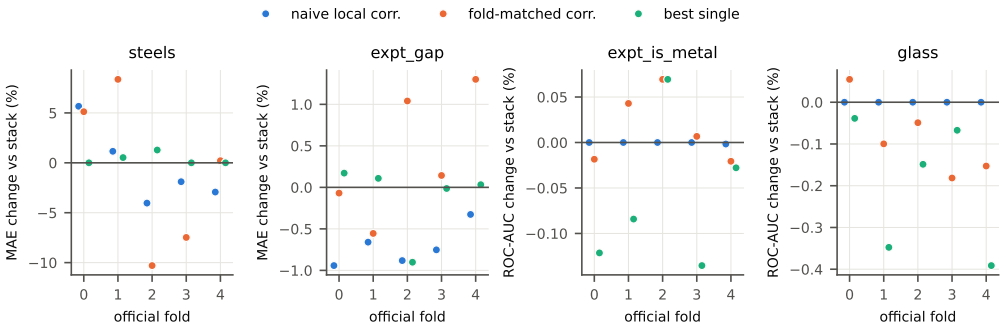


Figure 2: Per-fold change of the test metric relative to the uncorrected stack for the naive and fold-matched corrections and the best single expert (selected by inner CV). For classification a positive change in ROC-AUC is better.

Table 3: Number of official folds (of 5) in which the inner-CV selection switched each correction on, and the same for the label-permutation control.

Task	naive corr. on	fold-matched corr. on	perm. control on (naive)	perm. control on (f.-m.)
steels	5/5	5/5	5/5	5/5
expt_gap	5/5	5/5	5/5	5/5
expt_is_metal	1/5	5/5	5/5	5/5
glass	0/5	5/5	5/5	5/5

4.5. Comparison with the Matbench leaderboard

Table 4 places the pre-registered primary arm on the leaderboard snapshot of 18 October 2026. For the classification tasks we compare balanced accuracy at the inner-CV threshold with the board's balanced-accuracy column, because the board's "ROC-AUC" values of the label-based entries equal their balanced accuracy. The method would rank 3 of 12 on steels (behind TPOT-Mat and AutoML-Mat, ahead of MODNet v0.1.12), 2 of 13 on expt_gap (behind Darwin, ahead of the CrabNet and MODNet entries), 2 of 8 on expt_is_metal and 2 of 8 on glass. It does not beat the best entry on any task. On expt_gap and glass the gaps to the leaders are large (0.3240 vs. 0.2865 eV; balanced accuracy 0.8841 vs. 0.9603), and on expt_is_metal it is 0.023 in balanced accuracy. Our probability ROC-AUCs (0.9825 on expt_is_metal, 0.9567 on glass) are not comparable with the label-based board values and are not ranked.

Table 4: Primary arm versus the Matbench v0.1 leaderboard (snapshot 18 October 2026). Classification: balanced accuracy at the threshold chosen inside each training fold, compared with the board's balanced-accuracy column. Rank counts our entry among the board entries.

Task	Metric	Ours: stack + naive corr.	Best board entry	Rank
steels	MAE	82.80 ± 11.65	TPOT-Mat: 79.95 ± 13.59	3 of 12
expt_gap	MAE	0.3240 ± 0.0140	Darwin: 0.2865 ± 0.0083	2 of 13
expt_is_metal	balanced acc.	0.9370 ± 0.0063	Darwin: 0.9598	2 of 8
glass	balanced acc.	0.8841 ± 0.0184	MODNet (v0.1.12): 0.9603	2 of 8

Table 5: Calibration on the classification tasks (means over folds) and balanced accuracy at the inner-CV threshold.

Task	Arm	Brier	log loss	ECE	bal. acc.
expt_is_metal	Best single	0.0498	0.2007	0.0369	0.9378
expt_is_metal	Stack	0.0469	0.1662	0.0201	0.9374
expt_is_metal	Stack + naive corr.	0.0468	0.1662	0.0192	0.9370
expt_is_metal	Stack + fold-matched corr.	0.0469	0.1660	0.0211	0.9366
glass	Best single	0.0747	0.2451	0.0220	0.8827
glass	Stack	0.0735	0.2398	0.0211	0.8841
glass	Stack + naive corr.	0.0735	0.2398	0.0211	0.8841
glass	Stack + fold-matched corr.	0.0740	0.2411	0.0232	0.8835

4.6. Calibration and compute

Stacking improved the Brier score and the expected calibration error over the best single expert on both classification tasks, and the corrections left them essentially unchanged (Table 5). On 24 CPU cores (5 parallel workers of 4 threads) one official fold took a median of about 1, 16, 8 and 44 minutes for steels, expt_gap, expt_is_metal and glass when every expert was computed from scratch (the glass figure includes a system-standby pause); the pair-excluded refits of the fold-matched arm account for 44%, 33% and 37% of the expert-fitting time on the first three tasks (84% on glass, inflated by the same pause).

5. Results II: the strengthened pipeline against the leaderboard (Amendment 2)

5.1. Primary endpoint: final pipeline versus the board leader

Table 7 compares the frozen Amendment-2 pipeline with the leader of each per-task leaderboard. The comparison is paired over the same official

Table 6: Mean simplex-stack weight per expert (over folds; "–" where the expert was not in the selected configuration of every fold).

Expert	steels	expt_gap	expt_is_metal	glass
et_all	0.94	0.86	0.41	0.57
krr_frac	0.00	0.00	–	–
krr_magpie	0.06	0.00	–	–
lgbm_all	0.00	0.14	0.48	0.27
lgbm_l1_all	–	–	–	–
mlp_all	0.00	0.00	0.05	0.00
rf_all	0.00	0.00	0.03	0.10
svc_frac	–	–	0.01	0.06
svc_magpie	–	–	0.02	0.00

test compositions, using the leaders' public predictions.

expt_gap. The final pipeline reached an MAE of 0.2352 eV against 0.2865 eV for Darwin, a fine-tuned large language model and the current leader ($\Delta = -0.05131$ eV, 95% CI [-0.0701, -0.0331], $p_{\text{Holm}} < 0.001$). This is the only task where the pre-registered rule for "beats the leader" is met. It would place first among the 13 entries.

expt_is_metal. The balanced accuracy of 0.9547 is on par with Darwin's 0.9598 ($p_{\text{Holm}} = 0.299$). It would rank 2 of 8.

steels. The MAE of 80.10 MPa is on par with TPOT-Mat's 79.95 MPa (95% CI [-4.54, +4.89] MPa, $p_{\text{Holm}} = 0.928$). It would rank 2 of 12.

glass. The pipeline remains clearly behind MODNet v0.1.12: balanced accuracy 0.8937 versus 0.9603, $p_{\text{Holm}} < 0.001$. MODNet's probabilities, which are available in its submission, also give a much higher probability ROC-AUC (0.9899 versus our 0.9606).

5.2. What changed between the two pre-registered runs

Table 8 compares the two frozen pipelines on the board metric. The largest gain is on expt_gap, where the MAE fell from 0.3240 to 0.2352 eV. Two ingredients account for it. Both were identified on training rows only in the design phase (`results/design_a2/DESIGN_LOG_A2.md`):

- the DFT transfer block, which lowered the cross-validated MAE of a single LightGBM on fold 0's training rows from 0.374 to 0.302;

Table 7: Amendment-2 final pipeline versus the leaderboard leader of each task. The comparison is paired over the same official test compositions, using the leaders' public per-sample predictions. Metric semantics follow the board: MAE; balanced accuracy for classification (ours at the inner-CV threshold; Darwin's recorded labels; MODNet's probabilities at 0.5, as the board scored them). Δ is ours minus the leader. CI: paired bootstrap over test points within folds ($B = 10^4$). Holm correction over four tasks. Rank: among the board entries of 18 October 2026, including ours.

Task	Metric	Final pipeline	Leader (board)	Δ [95% CI]	p_{Holm}	Rank
steels	MAE	80.10 ± 10.19	TPOT-Mat: 79.95	+0.1577 [-4.54, +4.89]	0.928	2 of 12
expt_gap	MAE	0.2352 ± 0.0167	Darwin: 0.2865	-0.05131 [-0.0701, -0.0331]	<0.001	1 of 13
expt_is_metal	bal. acc.	0.9547 ± 0.0033	Darwin: 0.9598	-0.005093 [-0.0118, +0.00179]	0.299	2 of 8
glass	bal. acc.	0.8937 ± 0.0158	MODNet (v0.1.12): 0.9603	-0.06658 [-0.0759, -0.0573]	<0.001	2 of 8

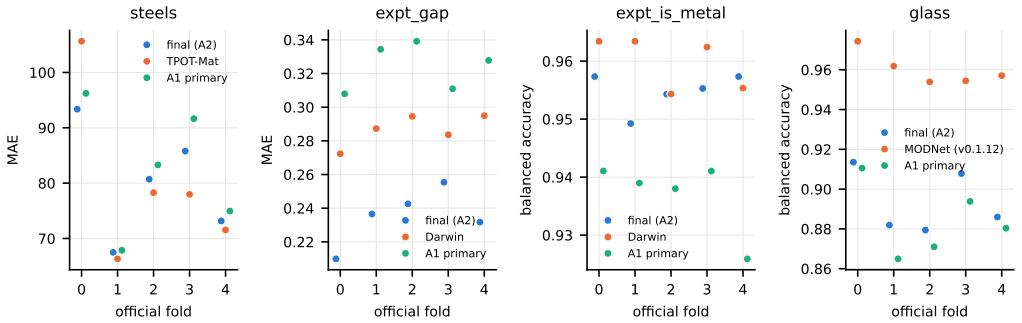


Figure 3: Per-fold board metric of the Amendment-2 final pipeline, the task leader and the Amendment-1 primary arm.

- hurdle experts, which treat the roughly half of compounds with $E_g = 0$ separately. They lowered it further to about 0.25.

A hurdle expert was the best single model in every fold (hurdle_lgbm_F2_median in most), and the stack added nothing on top of it (Δ vs. best single +0.08%, $p = 0.760$). On expt_is_metal the DFT block and the larger expert pool raised the balanced accuracy from 0.9370 to 0.9547. Glass improved modestly (0.8841 to 0.8937). Steels changed by -2.694 MPa (82.80 to 80.10 MPa, $p = 0.288$), which is within noise.

5.3. Stacking and the gated correction

Under the nested gate the local correction was kept in none of the 20 official folds (steels 0/5, expt_gap 0/5, expt_is_metal 0/5, glass 0/5). The final pipeline is therefore the simplex stack itself. This agrees with Amend-

Table 8: Board metric of the two pre-registered pipelines on the official folds, with a paired bootstrap p-value for the difference.

Task	Amendment 1 (primary arm)	Amendment 2 (final)	p_{boot}
steels	82.80	80.10	0.288
expt_gap	0.3240	0.2352	<0.001
expt_is_metal	0.9370	0.9547	<0.001
glass	0.8841	0.8937	<0.001

ment 1: once judged by an honest nested estimate, the local correction does not earn its place on these tasks.

The stack improved on the best single expert only on glass (+0.18% ROC-AUC, $p = 0.014$). It was statistically indistinguishable from it elsewhere. On steels the best single expert, extremely randomised trees on element fractions alone, reached a slightly lower MAE (79.56 MPa) than the stack (80.10 MPa). We report the pre-registered stack as the final result. Table 9 lists every arm and expert, and Table 10 the stack weights.

5.4. Compute

With five parallel workers of four threads on one 24-core workstation, one official fold took a median of about 9, 62, 48 and 57 minutes for steels, expt_gap, expt_is_metal and glass. The whole Amendment-2 run used 14.3 worker-hours. The Optuna tuning of LightGBM is the largest single cost.

6. Discussion

What the two pre-registered runs show. The first run tested the transfer of the novelty-damped local OOF residual correction from our chemistry study. It transferred only weakly: a consistent 0.7% MAE gain on the experimental band gap, nothing measurable on steels, and nothing on the classification tasks, where the inner selection switched it off. The second run strengthened the substrate instead. It added a disclosed DFT transfer block, a hurdle treatment of the zero-inflated band gap, more descriptors and more experts, and it gated the correction by a nested check. That gate then kept the correction in none of the 20 folds. The pipeline that emerges is therefore a well-regularised cross-fitted stack of standard composition experts. Its strength comes from *what the experts see* (DFT descriptors) and *what loss*

Table 9: Amendment-2 arms and experts on the official folds (mean ± s.d. over folds; probability ROC-AUC for classification).

Arm	steels MAE (MPa)	expt_gap MAE (eV)	expt_is_metal ROC-AUC	glass ROC-AUC
Best single (inner CV)	79.56 ± 9.90	0.2351 ± 0.0169	0.9900 ± 0.0010	0.9589 ± 0.0081
Equal-weight mean	95.18 ± 17.31	0.3172 ± 0.0112	0.9878 ± 0.0007	0.9524 ± 0.0095
Simplex stack	80.10 ± 10.19	0.2352 ± 0.0167	0.9895 ± 0.0008	0.9606 ± 0.0080
Final: stack + gated local corr.	80.10 ± 10.19	0.2352 ± 0.0167	0.9895 ± 0.0008	0.9606 ± 0.0080
Best single + gated corr.	79.56 ± 9.90	0.2351 ± 0.0169	0.9897 ± 0.0011	0.9589 ± 0.0081
Expert cat_F1	90.05	0.3075	0.9877	0.9475
Expert et_F1	84.47	0.2823	0.9856	0.9554
Expert et_F2	86.43	0.2819	0.9875	0.9600
Expert et_frac	79.56	–	0.9506	0.9476
Expert hurdle_et_F1_exp	–	0.2825	–	–
Expert hurdle_lgbm_F2_exp	–	0.2357	–	–
Expert hurdle_lgbm_F2_median	–	0.2360	–	–
Expert knn_frac	103.86	0.5019	0.9399	0.9078
Expert krr_frac	160.38	0.5613	–	–
Expert krr_magpie	109.37	0.4840	–	–
Expert lgbm_F2_tuned	88.38	0.2857	0.9900	0.9585
Expert mlp_F1	161.54	0.3950	0.9811	0.8937
Expert svc_frac	–	–	0.9472	0.8735
Expert svc_magpie	–	–	0.9566	0.9009
Expert xgb_F2	91.41	0.2902	0.9897	0.9571

structure they model (the point mass at $E_g = 0$), not from a local residual stage.

The experimental band gap. On expt_gap the final pipeline's MAE differs from the board leader's by -17.9% (0.2352 versus 0.2865 eV), and the margin is clear on a paired test. Two caveats apply. First, the gain relies on transfer from DFT calculations, which the leaders of this task do not use in the same form. We disclose it as transfer learning, use only the offline Materials Project data shipped with `matminer`, and use no experimental labels from outside the task. Second, 27% of the expt_gap compositions have no DFT match, and for them the pipeline falls back on composition descriptors. The hurdle construction is simple and general. A model that must predict exactly zero for half of the targets is poorly served by a single L1 or L2 regressor. Separating "is the gap non-zero?" from "how large is it?" removed a substantial part of that error in the design phase (MAE about 0.30 to 0.25 on fold 0's training rows).

The tasks where we do not lead. On expt_is_metal and steels the pipeline is statistically on par with the leaders (Darwin and TPOT-Mat), and on glass

Table 10: Mean Amendment-2 stack weight per expert over folds.

Expert	steels	expt_gap	expt_is_metal	glass
cat_F1	0.00	0.00	0.14	0.02
et_F1	0.00	0.00	0.00	0.03
et_F2	0.02	0.00	0.27	0.40
et_frac	0.85	–	0.01	0.24
hurdle_et_F1_exp	–	0.00	–	–
hurdle_lgbm_F2_exp	–	0.40	–	–
hurdle_lgbm_F2_median	–	0.60	–	–
knn_frac	0.03	0.00	0.01	0.00
krr_frac	0.00	0.00	–	–
krr_magpie	0.05	0.00	–	–
lgbm_F2_tuned	0.05	0.00	0.28	0.12
mlp_F1	0.00	0.00	0.07	0.00
svc_frac	–	–	0.00	0.00
svc_magpie	–	–	0.01	0.00
xgb_F2	0.00	0.00	0.21	0.19

it is clearly behind MODNet v0.1.12. MODNet's glass submission reaches a probability ROC-AUC of 0.9899, against our 0.9606. We could not close this gap with the descriptors developed for glass formation or with any CPU-feasible expert we tried. A MODNet expert itself exceeded our CPU budget. We did not audit the leaders' protocols. For example, we cannot tell from the public records whether all tuning was confined to the training folds. Our comparison is therefore against the reported numbers and predictions, not a claim about the methods.

Classification metrics on the leaderboard. The leaders of both classification tasks are scored by balanced accuracy, whatever the column name says. Darwin recorded hard labels. MODNet recorded probabilities, but the `matbench` version that produced the board entry thresholded them at 0.5 for every metric. A submission with probabilities scored by the current package would receive a true ROC-AUC (0.9895 for our expt_is_metal pipeline). That number is not comparable to the label-based entries. We recommend that future submissions and leaderboards report balanced accuracy and probability ROC-AUC as separate columns.

Limitations.

- **Scope.** The study covers four composition tasks and CPU-feasible experts only. Five folds give little power per task, except where effects are large.

- **Two looks at the test folds.** Amendment 2 was written after the Amendment-1 test results had been seen. Although every Amendment-2 choice was made on training rows only and the test folds were scored once under it, the decision to strengthen the pipeline, and the direction it took, were motivated by the first run's standing on the board. Both runs are reported in full.

- **Candidate configurations from fold 0.** The candidate configurations, feature blocks and expert pool were developed on fold 0's training rows. Those rows contain the test rows of folds 1–4, so for those folds the *space* of designs, though never their fitted parameters, was informed by data that later served as test data.

- **Optimistic tuning OOF.** The Optuna-tuned expert's OOF predictions are mildly optimistic, because tuning and stacking share inner splits. This can affect only the stack weights.

- **Dropped experts.** CrabNet- and MODNet-based experts were implemented but dropped for CPU cost, so the expert pool is not exhaustive.

7. Conclusions

We evaluated cross-fitted stacking of composition experts on the four Matbench v0.1 composition tasks under two pre-registered, fully nested protocols.

First run. A novelty-damped local OOF residual correction transferred only weakly from chemistry to materials.

Second run. The strengthened, still fully nested pipeline used DFT transfer descriptors and a hurdle model for the zero-inflated band gap. It:

- lowers the experimental band-gap MAE to 0.2352 eV, below the current leader (0.2865 eV);

- is on par with the leaders on expt_is_metal and steels;

- remains behind on glass.

We report all runs, folds and comparisons. Predictions and code are provided so that the numbers can be checked.

Data and code availability

All code, the frozen analysis plan (`ANALYSIS_PLAN.md`), SHA-256 freeze hashes, per-fold predictions of every arm, per-fold metrics and the unsubmitted Matbench result files are in the repository `https://github.com/bhaskargurram-ai/matbench-composition-stack`, which will be made public on submission. The data are the public Matbench v0.1 tasks.

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CRediT authorship contribution statement

Bhaskar Gurram: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic), running the ResearchStudio idea-generation skills, to assist with literature search, prior-art checking, code development and drafting of the manuscript. All analysis code, results and text were reviewed by the author, who takes full responsibility for the content of the publication.

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